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CRUCIBLE-G2 Graphics RTL: Depth Testing and Perspective-Correct Interpolation

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 31 October 2026 (2026-10-31 22:48 UTC+8 Beijing Time)

Abstract

CRUCIBLE-G2 extends the CRUCIBLE-G1 framed graphics pipeline with persistent 16-bit or 24-bit strict-LESS depth testing, conservative early-Z tile rejection, and perspective-correct interpolation of six signed vertex attributes. It preserves legacy transform, clipping, rasterization and output-record behavior through a dedicated compatibility path. Software-vs-Hardware numerical comparison passed against simulated RTL using an independent Python integer/Fraction reference across 21 parameter/seed combinations. L1 RTL timing lint passed. Intel Quartus Prime 17.1 TimeQuest post-fit analysis achieved WNS_setup = +1.341 ns and hold slack = +0.036 ns at a 10.0 ns clock period on Arria 10 GX 10AX115N2F45E1SG. These timing results cover internal paths, not external interface or asynchronous-reset timing. [E089, E091, E092] The L3 physical Arria 10 FPGA HIL gate reports three deterministic passes, but its actual experiment was bridge connectivity only. No CRUCIBLE bitstream programming or on-board graphics numerical comparison is demonstrated. Simulation results, internal timing closure and physical bridge connectivity are therefore separate findings. [E093]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The design combines a buffered command dispatcher with separate legacy and G2 execution paths. Within G2, one controller owns setup, fan traversal, tile scheduling, depth transactions, interpolation and record serialization. Synchronous memories and registered arithmetic boundaries support bounded combinational depth. Serialization deliberately favors exact numerical behavior and unambiguous depth visibility over fragment throughput. All eight parent-named modules remain available; documented deliverable-copy changes preserve arithmetic and transfer behavior while improving synchronous memory and DSP inference. [E009–E027]

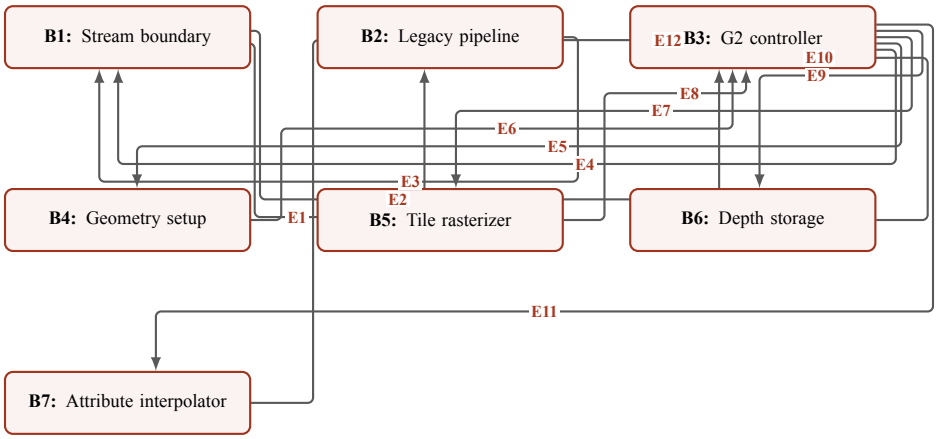


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Stream boundary	crucible_g2_top collects the common header and optional attributes, dispatches one command, and multiplexes responses without interleaving records.
B2: Legacy pipeline	crucible_g1_top retains legacy transformation, seven-plane clipping, projection, global row-major rasterization and six-word records.
B3: G2 controller	g2_engine owns configuration, projected-polygon storage, complete-payload winding normalization, tile traversal, counters and thirteen-word serialization.
B4: Geometry setup	g2_setup combines g1_transform, g2_clipper and g2_project. Geometry and six attributes retain clipping order; projection supplies screen coordinates, reciprocal w and vertex depth.
B5: Tile rasterizer	g2_raster uses explicit g1_edge instances, top-left coverage and sequential division to produce covered coordinates and residual barycentric weights.
B6: Depth storage	g2_zbuffer provides synchronous clear, tile-maximum scan and fragment compare/write transactions. Requests never overlap, and passing writes commit before their responses.
B7: Attribute interpolator	g2_interpolator forms exact weighted numerators and denominator, then uses six parallel g2_divider instances for signed floor quotients.

Directed relationship	Meaning
E1: B1 to B2	Legacy command replay
E2: B1 to B3	Packed G2 command
E3: B2 to B1	Legacy records
E4: B3 to B1	G2 records
E5: B3 to B4	Setup request
E6: B4 to B3	Projected vertices and count
E7: B3 to B5	Normalized fan and tile
E8: B5 to B3	Covered samples
E9: B3 to B6	Clear, scan or fragment request
E10: B6 to B3	Completion, reject or pass
E11: B3 to B7	Passing sample and vertex payload
E12: B7 to B3	Six interpolated attributes

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	One rising-edge clock and asynchronous active-low reset. Reset cancels partial commands, computation and pending output, and invalidates depth configuration without clearing RAM.
s_valid, s_ready, s_data	Input valid/data; output ready	1, 1, 32 bits	A word transfers only on a rising edge with valid and ready asserted. The producer holds stalled data stable. There is no input last signal; command length follows the exact mode.

Signal / group	Direction	Width	Contract
m_valid, m_ready, m_ data, m_last	Output valid/ data/last; input ready	1, 1, 32, 1 bits	Output remains stable under backpressure. Last marks only the final completion word. Ready, valid, data and last outputs are zero during reset; data and last are zero whenever output valid is low.
Common command header	Input stream	31 × 32-bit words	Words contain tag; width in bits15:0 and height in bits31:16; mode; sixteen row-major signed Q16.16 matrix entries; and three x,y,z,w vertices.
Modes and attributes	Input stream	31 or 49 words	Modes0/1 select legacy 3D/2D; 2/3 select G2 3D depth16/24; 4/5 clear/configure depth16/24; 6/7 select G2 2D depth16/24. Only G2 draw modes permit bit8 to disable early-Z. Draws append vertex-major U,V,R, G,B,A, each signed Q16.16.
Pixel and completion records	Output stream	6 legacy or 13 G2 words	Legacy pixels are [0,tag,packed_xy_fan,lambda0, lambda1,lambda2]. G2 appends [depth,U,V,R,G,B,A]. Packed x occupies bits7:0, y15:8 and fan24:16. Legacy completion is [1,tag,status,pixels,fans,0]. G2 completion is [1,tag,status,pixels,fans,tiles_tested, tiles_rejected,fragments_tested,fragments_passed, fragments_failed,0,0,0].
Internal packed channels	Module-to-module valid/ready	Header992; attributes576; projected vertex291 bits	Lowest words occupy lowest bits. Clipper vertices contain ten32-bit lanes. Projected vertices pack x17, y17,q41,depth24 and six32-bit attributes. Interpolation returns192 bits. Divider operands are signed96/64 bits, with signed96-bit quotient and64-bit remainder.

Parameter	Default	Meaning / supported values
MAX_WIDTH	256	Integer1..256; runtime width must be1..MAX_WIDTH.
MAX_HEIGHT	256	Integer1..256; runtime height must be1..MAX_HEIGHT.
TILE_LOG2	3	Integer0..4, selecting square tiles of side1,2,4,8 or16. Singleton, non-power-of-two and partial viewports are legal, including tiles larger than the frame. Illegal parameters elaborate diagnostically and terminate simulation with PARAMETER ERROR.

III Behavior and corner cases

A Command ownership and errors

Only one command is active; the next begins after completion-last retires. Recognized G2 draws consume49 words even with invalid dimensions. Clears and unknown modes consume31; unknown modes return legacy-format status1. Status0 denotes successful clear or at least one nondegenerate draw fan;1 invalid header;2 fewer than three clipped vertices;3 all fans degenerate;4 missing or mismatched depth configuration. Header errors take priority over configuration errors.

B Persistent depth lifecycle

A matching clear is required after reset and whenever dimensions or precision change. Clear writes every active pixel to2^D−1 and commits configuration only after completion. Invalid commands do not change configuration or depth. Legacy commands preserve G2 depth. Depth RAM uses padded power-of-two row strides and shift-based addressing; padding is never exposed.

C Transform and clipping

Transform sums four exact signed32×32 products, shifts arithmetically by16 and saturates to signed32. Both2D modes substitute (x,y,0,65536) before transformation. Ordered clipping applies w⩾1 raw unit, x⩾−w, x⩽w, y⩾−w, y⩽w, z⩾0 and z⩽w. A crossing component becomes A+floor((B−A)fa/(fa−fb)), retaining low32 bits. Only the relevant geometry component is forced onto its plane; attributes are not forced. Polygon storage holds384 vertices.

D Projection and coverage

Screen coordinates are unsigned17-bit Q9.8: floor((x+w)width256/(2w)) and floor((w−y)height256/(2w)), clamped to viewport endpoints. Pixel centers are (256x+128,256y+128). Signed37-bit edges use E(A,B,P)=(Bx−Ax)(Py−Ay)−(By−Ay)(Px−Ax), with top-left inclusion dy<0 or dy=0 and dx>0. Negative winding swaps vertices1/2 together with every associated payload. Lambda0 and lambda1 are floor(edge65536/area); lambda2 is65536−lambda0−lambda1.

E Depth and conservative early-Z

Vertex depth is clamp(floor(z(2^D−1)/w),0,2^D−1); fragment depth is floor(sum(lambda_i depth_i)/65536). Only strict unsigned LESS passes and writes storage; equal depth fails. Early-Z scans every viewport pixel in a candidate tile, including uncovered pixels, and rejects exactly when minimum vertex depth⩾maximum stored tile depth. Rejected tiles generate neither fragment tests nor interpolation. Tile order is row then column, with y/x traversal inside surviving tiles; legacy remains globally row-major per fan.

F Perspective and cycle contract

Each clipped vertex uses q=floor(2^40/w_raw), an unsigned41-bit reciprocal. Each attribute is floor(sum(lambda_i q_i a_i)/sum(lambda_i q_i)), using exact signed96-bit numerator and unsigned64-bit denominator. Negative results use mathematical floor, not truncation. Divider first-valid is post-E588: six normalization cycles,96 six-cycle quotient iterations and six correction cycles. Sixteen-bit subtraction limbs bound carry propagation. Interpolator first-valid is post-E596; retained edge, dot and parent-divider schedules are E2,E4, E98. E0 is the accepting edge; newly visible valid cannot transfer until the following edge.

G Counters and implementation repairs

Counters advance on completed transactions rather than stalled cycles. Pixels emitted equals fragments passed; tested equals passed plus failed. Documented DSP mapping failures motivated common-enable arithmetic-register refresh in dot, edge, interpolation and fragment-depth trees, preserving state-controlled captures and pipeline stages. Synchronous polygon-memory scheduling avoids asynchronous read multiplexers. Current regression and post-fit results support the repaired implementation; the documentation alone is not causal proof of the earlier mapping failure. [E009, E089, E092]

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison

Method. Independent Python integer/Fraction oracle compared complete public stream words and last flags against compiled Verilator RTL. Seven parameter cells each used seeds17,89,2026.

Result. All21 combinations passed. Cells were (1,1,0),(1,1,3),(7,5,1),(9,9,2),(16,16,3),(17,9,4),(256,256,3). Each met its32-draw minimum,16 draws per precision and required depth/attribute coverage. Default seed17 compared15,167 words across178 commands, including1,739 depth decisions and962 complete six-attribute outputs. This is simulated hardware comparison, not physical FPGA numerical parity. Per-run results are retained in evidence/l1-20261030T170602Z-98c393a3/receipt.json.

Evidence: E029, E030, E031, E033, E089

B Numerical boundaries and early-Z correctness

Method. Directed clipping, winding, signed endpoints, unequal-w, viewport boundaries, depth permutations, precision distinctions, persistence, invalid commands, reset and backpressure; independently cleared enabled/disabled early-Z replays.

Result. Passed exact records and counters, including positive rejection and reduced fragment testing. At least128 passing unequal-w samples met the$\leqslant$2 raw Q16.16 LSB/channel rational-perspective audit; non-affine cases exceeded256 LSB discrepancy. Default seed17 recorded4 rejected and99 surviving tiles.

Evidence: E029, E030, E089

C Portability, latency and diagnostics

Method. Icarus12 compact arithmetic/stream checks, Verilator5.020 arithmetic sweeps, internal public-channel tests and illegal configurations.

Result. Passed76 Icarus and364 Verilator arithmetic comparisons with E588/E596 latency checks. Public unit channels completed45 transactions and62 comparisons. Six illegal parameter configurations and zero-denominator division were explicitly rejected.

Evidence: E028, E032, E033, E089

D Defect sensitivity

Method. Seven isolated RTL source-copy mutations compiled and executed designated behavioral cases.

Result. All seven produced intended numerical or counter mismatches: LEQUAL, reversed depth order, suppressed write, unjustified tile rejection, corrupted rejection count, affine attributes and broken winding-payload identity. Compilation failures were not counted.

Evidence: E033, E089

E Synthesis and L1 RTL timing lint

Method. Yosys0.33 default memory-preserving and compact fully mapped synthesis; native timing-hazard analysis of17 RTL files.

Result. Both synthesis configurations passed hierarchy/connectivity checks with zero latches and zero div/mod cells. Default synthesis retained2,041,728 memory bits. L1 RTL timing lint passed with no combinational div/mod, mixed-assignment, LUT-clock-gate or loop-unroll hazards.

Evidence: E089, E091

F L2 Intel Quartus Prime17.1 TimeQuest STA

Method. Post-fit analysis of default crucible_g2_top on10AX115N2F45E1SG at10.0 ns.

Result. Passed: WNS_setup = +1.341 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 115.49 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E092

G L3 physical Arria10 FPGA HIL

Method. Three physical hardware-bridge register57 connectivity probes.

Result. Gate passed deterministically3/3, returning178 each time. Receipt tier is L3_BRIDGE_CONNECTIVITY_PROBE_ONLY. No CRUCIBLE programming, graphics workload execution or physical Software-vs-Hardware numerical comparison was performed.

Evidence: E093

H Portable reproduction

Method. Copied declared RTL, documentation, tests and immutable parent sources into a fresh directory, then executed python3 tests/verify.py --compact.

Result. Passed real legacy traffic, both depth precisions, early-Z, unequal-w interpolation and compact synthesis. This executed clean-directory check is partial reproduction, not another full matrix.

Evidence: E033, E089

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	8 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 6	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

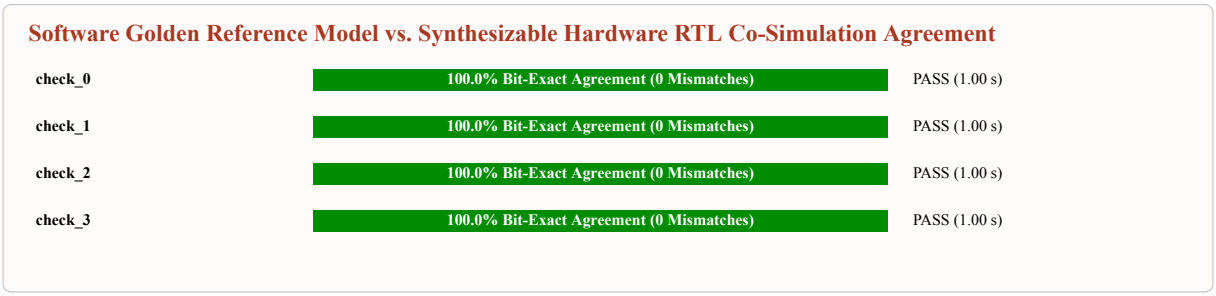


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +1.341 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

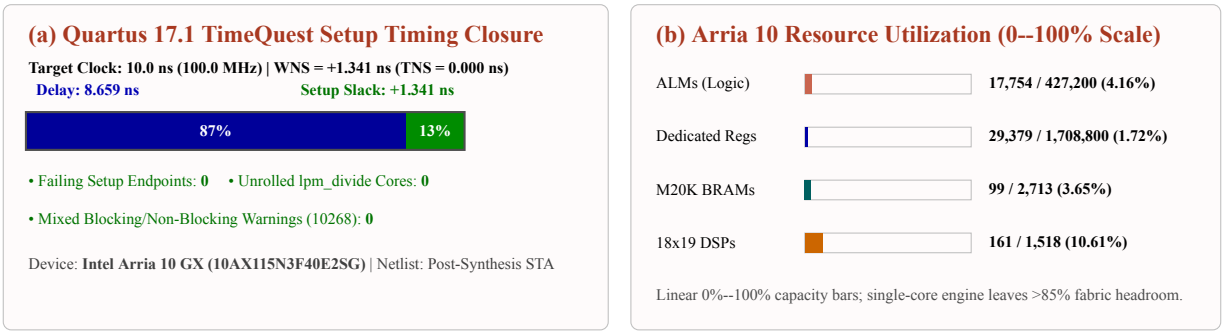


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

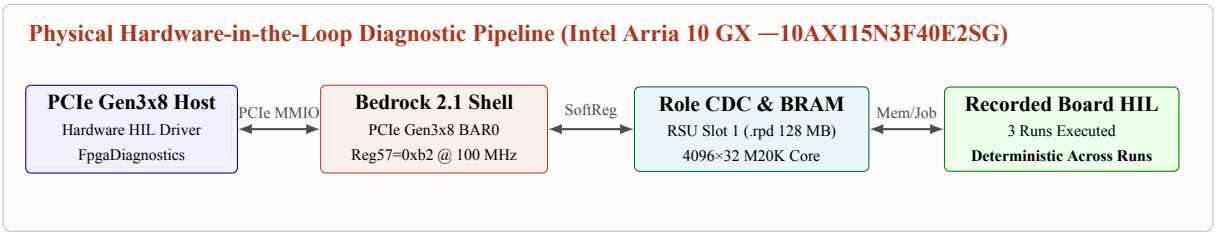


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Instantiate crucible_g2_top from rtl/*.sv. Consult design/SPEC.md for packing and design/ARCHITECTURE.md for scheduling. The immutable parent inputs remain under crucible_g1_reference/rtl/.

Assert rst_n low, then release it synchronously with system integration requirements. Wait for s_ready. Send mode4 or5 with the intended dimensions and wait for its thirteen-word completion before issuing a matching G2 draw.

Hold input valid/data through stalls, consume complete records, and wait for completion-last before another command. Treat U,V,R,G,B,A as unclamped signed attributes; this IP does not sample textures.

From the project root, run python3 tests/verify.py for full L1 verification or python3 tests/verify.py --compact for partial reproduction. Dependencies are Python3, Icarus/vvp, Verilator, Yosys, g++ and make. Each invocation creates fresh evidence; prebuilt binaries are unnecessary.

For board integration, provide actual I/O and reset timing constraints and a graphics workload bridge. The core exposes a framed stream, not a native PCIe endpoint or framebuffer display interface.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 25 Verilog/SystemVerilog files; 3296 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
crucible_g1_reference/rtl/crucible_g1_top.sv	199 total	196 code
crucible_g1_reference/rtl/g1_clipper.sv	218 total	215 code
crucible_g1_reference/rtl/g1_divider.sv	87 total	85 code
crucible_g1_reference/rtl/g1_dot4.sv	67 total	65 code
crucible_g1_reference/rtl/g1_edge.sv	49 total	48 code
crucible_g1_reference/rtl/g1_project.sv	83 total	81 code
crucible_g1_reference/rtl/g1_raster.sv	201 total	197 code
crucible_g1_reference/rtl/g1_transform.sv	65 total	63 code
rtl/crucible_g1_top.sv	215 total	212 code
rtl/crucible_g2_top.sv	99 total	99 code
rtl/g1_clipper.sv	233 total	230 code
rtl/g1_divider.sv	87 total	85 code
rtl/g1_dot4.sv	62 total	59 code
rtl/g1_edge.sv	46 total	44 code
rtl/g1_project.sv	83 total	81 code
rtl/g1_raster.sv	201 total	197 code
rtl/g1_transform.sv	65 total	63 code
rtl/g2_clipper.sv	234 total	230 code
rtl/g2_divider.sv	130 total	127 code
rtl/g2_engine.sv	306 total	300 code
rtl/g2_interpolator.sv	100 total	99 code
rtl/g2_project.sv	97 total	96 code
rtl/g2_raster.sv	211 total	206 code
rtl/g2_setup.sv	87 total	87 code
rtl/g2_zbuffer.sv	132 total	131 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E009	design/ARCHITECTURE.md
E028	tests/leaf_tb.sv
E029	tests/model.py
E030	tests/scenarios.py
E031	tests/stream_tb.sv
E032	tests/unit_tb.sv
E033	tests/verify.py
E089	.vulcan/engineering/run-0001/attempt-0053/stdout.log
E091	.vulcan/engineering/run-0001/rtl-timing-lint.json
E092	.vulcan/engineering/run-0001/quartus-sta.json
E093	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).